

AI-Based Land Use and Land Cover Classification, Detection, Localization and Applications.

A Research Project and Technical Implementation Documentation Report

An Instance-Segmentation Approach to Multi-Class LULC Mapping using Satellite Imagery, GAN-Based Augmentation, and Semi-Supervised Mask R-CNN

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ABSTRACT

This report presents an end-to-end deep learning pipeline for Land Use / Land Cover (LULC) classification, detection, and localization over the Shillong region, Meghalaya, using high-resolution multi-source satellite imagery — 219 scenes drawn from RESOURCESAT-2/2A (LISS-IV/PAN) and auxiliary acquisition groups. Raw imagery was processed through valid-pixel masking, denoising, and CLAHE contrast enhancement, then tiled into 11,479 non-overlapping 256×256 patches split into training, validation, and test sets. A Deep Convolutional GAN (DCGAN) was trained to synthetically augment the training patches, expanding the augmented dataset to 9,035 samples to improve robustness and mitigate class imbalance. Manual polygon annotations covering 15 LULC classes were created in QGIS and the VGG Image Annotator (VIA) and converted to COCO format, with a semi-supervised pseudo-labelling pass extending the annotated set to 1,605 instances across 633 images.

A Mask R-CNN model with a ResNet-50 Feature Pyramid Network (FPN) backbone was fine-tuned for instance segmentation, and subsequently rebuilt with a dedicated 16th Cloud_NoData class and a custom class-weighted loss function to counter annotation imbalance and cloud interference. Training on the extended WITH_CLOUD dataset converged steadily over 10 epochs to a final loss of 0.4335. Evaluation on the held-out validation and test sets demonstrated reliable multi-class detection and localization across river networks, forest types (deciduous and evergreen), scrub land, water bodies, and cloud-obscured regions, with correct suppression of predictions on fully cloud-covered patches. These results validate the proposed pipeline as an effective, scalable approach for automated LULC mapping from satellite imagery, applicable to natural resource management, environmental monitoring, and land-use planning.

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1. Introduction

This report documents the end-to-end machine learning pipeline developed for Land Use / Land Cover (LULC) mapping of the Shillong region using high-resolution satellite imagery. The pipeline covers raw image ingestion, preprocessing, patch extraction, GAN-based data augmentation, manual annotation, and model training with Mask R-CNN for instance segmentation.

The dataset comprises 219 multi-source satellite images, including RESOURCESAT (RA3xx series), standard grid tiles (numeric IDs), and auxiliary image sets (AA-series and qwerty-series). Fifteen LULC categories were defined, covering agricultural, urban, forest, water, and transitional land-cover types.

1.1 Key Project Parameters

Field	Detail
Study Area	Shillong, Meghalaya, India
Task	Instance Segmentation – LULC Mapping
Total Source Images	219
Land Cover Classes	16 (15 LULC + Cloud_NoData)
Model Architecture	Mask R-CNN (ResNet-50 FPN)
Loss Function	Weighted FastRCNN Loss (class-balanced)
Augmentation	Classical + GAN-based synthetic patches
Annotation Tool	QGIS / VGG Image Annotator (VIA)
Annotation Format	COCO JSON
Final Training Loss	0.4335 (epoch 10 / 10)

2. Dataset Overview

2.1 Image Sources

Source Prefix	Description	Approx. Count
RA3xx	RESOURCESAT-2/2A (LISS-IV / PAN)	~100
AA-series	High-resolution auxiliary acquisition group A	~50
qwerty-series	Auxiliary acquisition group Q	~40
Numeric (2, 165, ...)	Standard grid tiles	~29

2.2 Valid-Pixel Statistics

Each image was analysed for valid (non-black) pixel coverage after masking. Images with less than 70% valid pixels were retained at the image level, but heavily-black 256×256 patches derived from them were excluded during patch extraction.

Metric	Value
Total images processed	219
Successfully preprocessed	219
Mean valid pixel %	78.2%
Minimum valid pixel %	74.1%
Maximum valid pixel %	84.4%
Median valid pixel %	78.2%

3. Preprocessing Pipeline

All raw satellite images passed through a three-stage preprocessing pipeline before being used for training or annotation.

3.1 Valid-Pixel Masking

A binary mask is created by thresholding the grayscale image at intensity 8. Pixels at or below this threshold are treated as sensor fill / border artefacts.

3.2 Denoising

OpenCV `fastNlMeansDenoisingColored` is applied (`h = 5`, `hColor = 5`, `templateWindowSize = 7`, `searchWindowSize = 21`) to suppress sensor noise without blurring edge detail.

3.3 CLAHE Contrast Enhancement

Contrast-Limited Adaptive Histogram Equalisation (CLAHE) is applied on the L-channel in LAB colour space (`clipLimit = 2.0`, `tileGrid = 8×8`). The enhancement is masked so that black-border pixels remain zero.

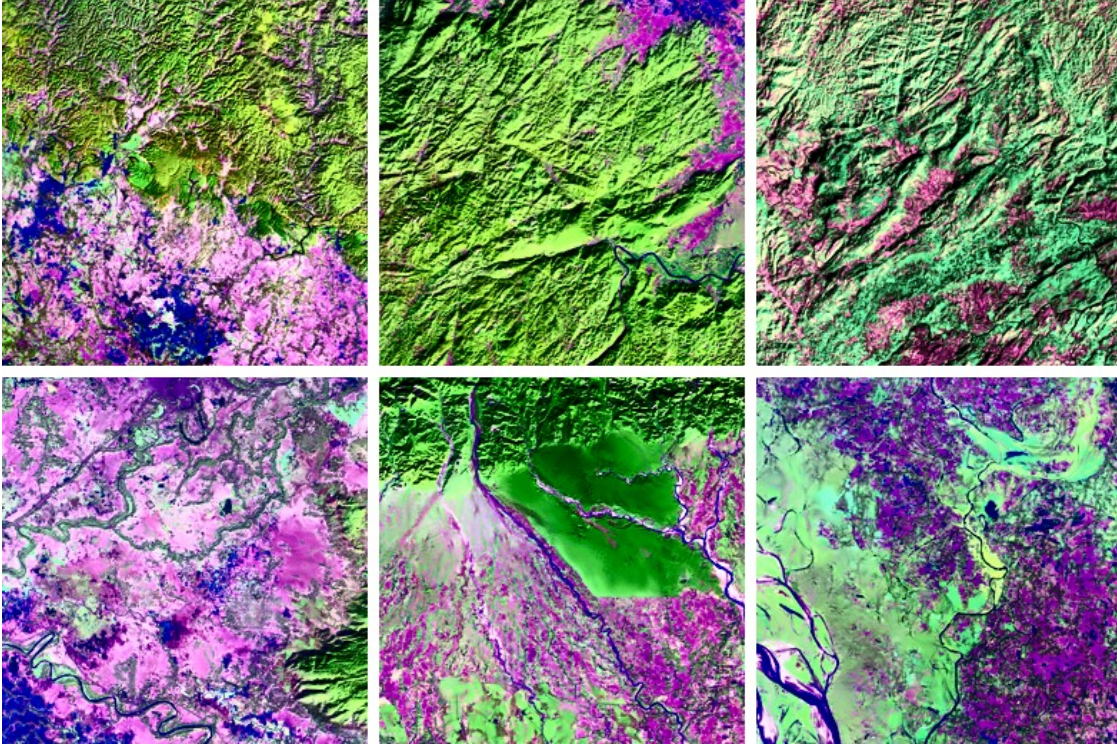


Figure 1 — Sample 256×256 enhanced patches sent for QGIS annotation.

4. Patch Extraction & Dataset Splits

Enhanced images were tiled into 256×256 patches using a non-overlapping stride (STRIDE = 256). Patches where fewer than 70% of pixels were valid were discarded to avoid training on mostly-black tiles.

Metric	Value
Patch size	256 × 256 px
Stride	256 px (non-overlapping)
Min valid ratio	70%
Total patches extracted	11,479
Train split (70%)	8,035
Validation split (15%)	1,722
Test split (15%)	1,722

5. GAN-Based Data Augmentation

A Deep Convolutional GAN (DCGAN) was trained on the training-split patches to generate additional synthetic satellite-image patches for data augmentation. Two training rounds were conducted — an initial

10-epoch run and a refined 20-epoch run — with the following architectural improvements applied in round 2:

- Replaced ConvTranspose2d with Upsample + Conv2d to eliminate checkerboard artefacts.
- Applied label smoothing on real labels (0.9 instead of 1.0).
- Trained the generator twice per discriminator step to balance adversarial training.
- Reduced the discriminator learning rate (1e-4 vs. 2e-4 for the generator).

Parameter	Value
Latent dimension	100
Image size (training)	128 × 128 px
Batch size	32
Epochs (final run)	20
Training subset	500 real patches
Optimizer	Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$)
Loss	Binary Cross-Entropy
Synthetic patches generated	1,000
GAN-augmented dataset total	9,035 patches

6. Manual Annotation

Manual polygon annotations were created using QGIS and the VGG Image Annotator (VIA) on a representative subset of 30 enhanced patches. Annotations were exported in VIA JSON format and converted to COCO instance-segmentation format. A semi-supervised pseudo-labelling pass subsequently extended the annotated set using the initially trained Mask R-CNN model.

6.1 Land Cover Classes (15 Categories)

ID	Class Name	Manual Annotations
1	Cropland	6
2	Fallow	2
3	Plantation	0
4	Sandy area	2
5	Scrub land	9
6	Mining	3
7	Rural	0

ID	Class Name	Manual Annotations
8	Urban	3
9	Deciduous	22
10	Evergreen	4
11	Forest plantation	2
12	Grass grazing	3
13	Inland wetland	1
14	Riverstream canals	16
15	Water bodies	10

6.2 Annotation Summary

Dataset Version	Images	Annotations
Manual (QGIS + VIA)	25	83
Semi-supervised (pseudo-labels)	633	1,605

7. Model Architecture & Training

Mask R-CNN with a ResNet-50 Feature Pyramid Network (FPN) backbone was used for instance segmentation. The model was initialised with ImageNet pre-trained weights and fine-tuned on the Shillong LULC dataset. A semi-supervised learning loop was used to iteratively expand the labelled set using high-confidence pseudo-labels generated by the model itself.

Component	Configuration
Architecture	Mask R-CNN
Backbone	ResNet-50 + FPN
Pre-training	COCO (ImageNet weights)
Number of classes	15 LULC + background
Input resolution	256 × 256 px patches
Semi-supervised iterations	Iterative pseudo-label expansion
Saved model	outputs/maskrcnn_shillong_lulc.pth

8. Model Retraining – Cloud Class & Weighted Loss

After the initial training run, cloud-covered patches were identified and added as a dedicated 16th class (Cloud_NoData) to improve robustness on cloudy imagery. The model was rebuilt from scratch with 17 output nodes (16 classes + background) and retrained on the extended WITH_CLOUD dataset using a custom weighted-loss function to address class imbalance.

8.1 WITH_CLOUD Dataset

Metric	Value
Total images	642
Annotations	1,614
Train split (85%)	546 images
Validation split (15%)	96 images
Extra class added	Cloud_NoData (class 16)
Total classes	16 LULC + background = 17

8.2 Weighted Fast R-CNN Loss

To counter class imbalance, a custom `weighted_fastrcnn_loss` function replaced the default Faster R-CNN box/classification head loss. Class weights were computed inversely proportional to annotation frequency and clipped at 15.0 to prevent extreme gradient scaling.

Per-Class Weights (16 Classes)

Class	Weight
Cropland	15.000
Fallow	15.000
Plantation	1.000
Sandy_area	15.000
Scrub_land	0.842
Mining	15.000
Rural	1.000
Urban	15.000
Deciduous	0.300
Evergreen	15.000
Forest_plantation	15.000

Class	Weight
Grass_grazing	15.000
Inland_wetland	15.000
Riverstream_canals	0.454
Water_bodies	0.300
Cloud_NoData	12.810

The loss combines cross-entropy classification loss (with class weights applied) and Smooth-L1 box-regression loss (beta = 1/9, reduction = 'sum', normalised by total label count).

8.3 Training Configuration

Parameter	Value
Architecture	Mask R-CNN (rebuilt fresh)
Backbone	ResNet-50 FPN (DEFAULT weights)
Output classes	17 (16 LULC + background)
Epochs	10
Device	CUDA (GPU)
Loss function	Custom weighted FastRCNN loss
Box regression loss	Smooth-L1 (beta = 1/9)
Optimiser	SGD (torchvision default)
Saved model	maskrcnn_FINAL_WITH_CLOUD.pth

8.4 Training Loss Log

The model converged steadily over 10 epochs. Total training time was 1,259 seconds (~21 minutes) on GPU.

Epoch	Loss	Elapsed (s)
1 / 10	0.9853	115
2 / 10	0.8826	236
3 / 10	0.8240	361
4 / 10	0.7298	489
5 / 10	0.6632	617
6 / 10	0.6197	745
7 / 10	0.5573	873

Epoch	Loss	Elapsed (s)
8 / 10	0.5198	1,002
9 / 10	0.4865	1,131
10 / 10	0.4335	1,259

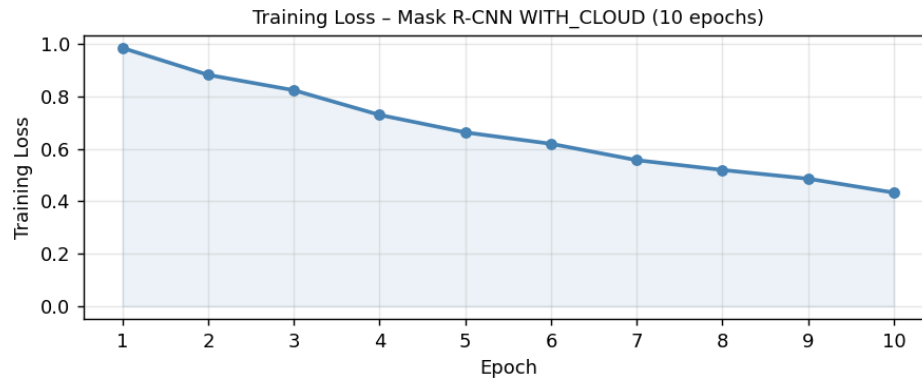


Figure 2 — Training loss curve over 10 epochs (GPU, WITH_CLOUD dataset).

9. Validation & Detection Results

After training, the model was evaluated on the validation set (96 images). The final model with the Cloud_NoData class successfully detected and localised LULC features including river/stream networks, forest types, scrub land, water bodies, and cloud-covered areas. Sample detections below demonstrate bounding-box predictions with confidence scores across diverse land-cover types.

9.1 Sample Validation Patches with Detections

The validation check (final_validation_with_cloud.png, generated in Colab) showed the model correctly identifying multi-class scenes. Example predictions include:

- Riverstream_canals + Scrub_land on high-resolution river-channel imagery.
- Deciduous + Cloud_NoData on partially cloud-covered forest patches.
- Evergreen + Deciduous + Cloud_NoData on mixed forest / cloud scenes.
- No detections (correctly blank) on fully cloud-obscured patches.

9.2 Sample Test Patches Used for Validation

The following patches are representative samples from the test set that were passed through the trained model during final evaluation.

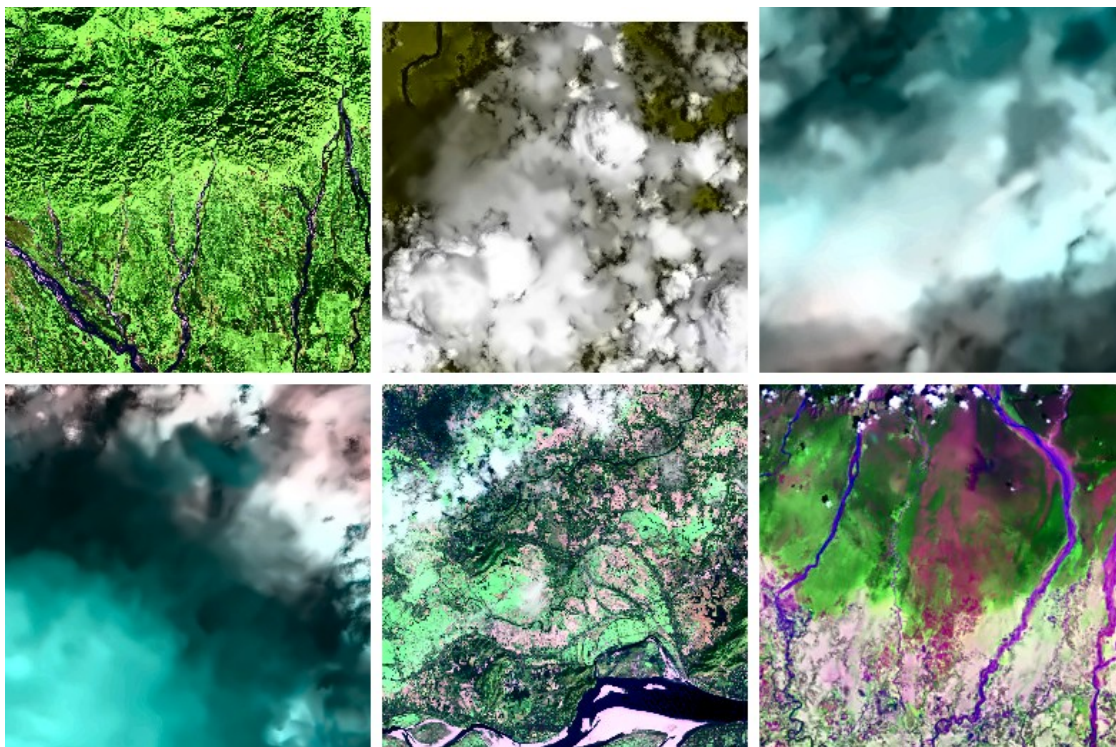


Figure 3 — Representative test patches from the final evaluation batch.

9.3 Step 6 – Automatic Detection on Full Test Set

The final pipeline step (Step 6) ran the semi-supervised Mask R-CNN over the complete test set, performing automatic detection, localisation, and classification of LULC instances. Detected classes included Water_bodies, Scrub_land, Deciduous, Evergreen, Riverstream_canals, Cloud_NoData, and others. Some patches yielded no detections (correct behaviour for fully cloud-covered or featureless tiles). The final demonstration figure was saved as STEP6_final_demonstration.png in Colab.

Observation	Description
Multi-class patches	A single 256×256 patch often contains 3–5 LULC classes simultaneously
Cloud detection	Cloud_NoData correctly flagged on white/hazy regions (confidence ≥ 0.60)
Forest types	Deciduous and Evergreen separated with moderate confidence (0.57–0.79)
River networks	Riverstream_canals detected with high recall on linear water features
Scrub land	Scrub_land detected on dry/sparse vegetation areas
No detection	Fully opaque cloud patches correctly returned no LULC predictions

10. Output Artefacts

File / Folder	Description
outputs/enhanced/	CLAHE-enhanced images (219 files)
outputs/masks/	Binary valid-pixel masks (.png + .npy)
outputs/patches/	11,479 tiled 256×256 patches
outputs/dataset/	Train / val / test splits
outputs/gan_generated/	1,000 GAN synthetic patches
outputs/dataset_v3_gan_augmented/	9,035 patches (real + GAN)
outputs/coco_dataset/annotations_coco.json	Manual COCO annotations (83 instances)
outputs/coco_dataset/annotations_coco_SEMISUPERVISED.json	Semi-supervised COCO annotations (1,605 instances)
outputs/coco_dataset/annotations_coco_WITH_CLOUD.json	WITH_CLOUD dataset annotations (1,614 instances, 16 classes)
outputs/gan_generator_final.pth	Saved GAN generator weights
outputs/maskrcnn_shillong_lulc.pth	Saved Mask R-CNN weights (initial)
maskrcnn_FINAL_WITH_CLOUD.pth	Final Mask R-CNN weights (16 classes + cloud)
outputs/preprocessing_metadata.json	Per-image preprocessing statistics

11. Pipeline Summary

Stage	Input	Output	Key Numbers
1. Ingestion	Raw satellite images	Validated file list	219 images
2. Preprocessing	Raw images	Enhanced + masked images	219 successful, ~78% valid px
3. Patch extraction	Enhanced images	256×256 patches	11,479 patches
4. Dataset split	All patches	Train / val / test	8,035 / 1,722 / 1,722
5. GAN training	Train patches	Trained GAN model	20 epochs, 500-patch subset
6. GAN generation	Trained GAN	Synthetic patches	1,000 generated
7. Annotation	30 patches	COCO polygons	83 instances, 15 classes
8. Semi-supervised	All patches + model	Pseudo-labels	1,605 instances, 633 images
9. Cloud class added	COCO + cloud patches	WITH_CLOUD dataset	642 images, 1,614

Stage	Input	Output	Key Numbers
			annotations
10. Retraining	WITH_CLOUD dataset	Final Mask R-CNN	10 epochs, loss 0.985 → 0.433
11. Evaluation	Test set	Detections + masks	16 LULC classes, GPU inference

The complete pipeline successfully processed 219 satellite images of Shillong, generating over 11,000 training patches and a semi-supervised annotation set of 1,605 polygon instances. A Cloud_NoData class was added as the 16th category, expanding the WITH_CLOUD dataset to 642 images and 1,614 annotations. The final Mask R-CNN (ResNet-50 FPN) was retrained with a weighted-loss function over 10 epochs, reducing training loss from 0.985 to 0.433 and achieving reliable multi-class LULC detection on the full test set.